Problem Set 5 Solutions: Diffusion and Criticality

1. Infinite Multiplication in a Graphite Pile

(a) **Calculate η:** The reproduction factor η is the number of fission neutrons produced per absorption in the *fuel*. For Natural Uranium, we must account for the absorption in both ^{235}U and ^{238}U.

$$\eta = \nu \frac{\Sigma_f^{235}}{\Sigma_a^{235} + \Sigma_a^{238}} = \nu \frac{N_{235}\sigma_f^{235}}{N_{235}\sigma_a^{235} + N_{238}\sigma_a^{238}}$$

Using atom fractions for Natural Uranium ($f_{235} = 0.0072$, $f_{238} = 0.9928$):

$$\begin{aligned}
\text{Numerator} &= 2.42 \times (0.0072 \times 582) = 10.14 \\
\text{Denominator} &= (0.0072 \times 680) + (0.9928 \times 2.7) \\
&= 4.896 + 2.68 = 7.576 \\
\eta &= \frac{10.14}{7.576} \approx \mathbf{1.338}
\end{aligned}$$

(Note: If you neglected ^{238}U absorption entirely, you would get $\eta \approx 2.07$, which is far too high for natural uranium.)

(b) **Calculate k_∞:** Using the Four Factor Formula:

$$k_\infty = \epsilon p f \eta$$

$$k_\infty = (1.03)(0.89)(0.88)(1.338)$$
$$k_\infty = 1.079$$

(c) **Conclusion:** Since $k_\infty > 1$, the lattice is capable of sustaining a chain reaction if the reactor is made large enough (infinite size).

2. Calculating Transport Properties

(a) **Transport Mean Free Path:**

$$\lambda_{tr} = \frac{1}{\Sigma_{tr}} = \frac{1}{0.45 \text{ cm}^{-1}} = \mathbf{2.22} \text{ cm}$$

(b) **Diffusion Coefficient:**

$$D = \frac{\lambda_{tr}}{3} = \frac{2.22}{3} = \mathbf{0.741} \text{ cm}$$

(c) **Diffusion Length:**

$$L^2 = \frac{D}{\Sigma_a} = \frac{0.741}{0.005} = 148.2 \text{ cm}^2$$
$$L = \sqrt{148.2} = \mathbf{12.17} \text{ cm}$$

3. Geometric Buckling and Critical Dimensions

(a) **Material Buckling (B_m^2):** Using $k_\infty = 1.05$ and $L^2 = 148.2$ cm^2:

$$B_m^2 = \frac{k_\infty - 1}{L^2} = \frac{1.05 - 1}{148.2} = \frac{0.05}{148.2} = \mathbf{3.374 \times 10^{-4}} \text{ cm}^{\mathbf{-2}}$$

$$B_m = \sqrt{3.374 \times 10^{-4}} = 0.01837 \text{ cm}^{-1}$$

(b) **Critical Sphere Radius (R_c):** For a sphere, $B_g^2 = (\frac{\pi}{R})^2$. Set $B_g^2 = B_m^2$:

$$\frac{\pi}{R_c} = B_m \implies R_c = \frac{\pi}{0.01837} = 171.0 \text{ cm} = \mathbf{1.71} \text{ m}$$

(c) **Critical Cube Side (a_c):** For a cube, $B_g^2 = 3(\frac{\pi}{a})^2$.

$$\sqrt{3}\frac{\pi}{a_c} = B_m \implies a_c = \frac{\sqrt{3}\pi}{0.01837} = \sqrt{3}(171.0) = 296.2 \text{ cm} = \mathbf{2.96} \text{ m}$$

(d) **Volume Comparison:**

$$V_{sphere} = \frac{4}{3}\pi R^3 = \frac{4}{3}\pi(1.71)^3 \approx \mathbf{20.95} \text{ m}^{\mathbf{3}}$$

$$V_{cube} = a^3 = (2.96)^3 \approx \mathbf{25.93} \text{ m}^{\mathbf{3}}$$

Conclusion: The sphere is more efficient (requires less volume/fuel) because it minimizes the surface-area-to-volume ratio, thereby minimizing neutron leakage.

4. The Effect of a Reflector

(a) **New Physical Radius:** The Reflector Savings δ essentially "replaces" part of the fuel. The extrapolated (bare) size remains constant, but the physical fuel radius decreases.

$$R_{physical} = R_{bare} - \delta$$

$$R_{physical} = 2.5 \text{ m} - 0.25 \text{ m} = \mathbf{2.25} \text{ m}$$

(b) **Volume Reduction:**

$$V_{old} \propto (2.5)^3 = 15.625$$

$$V_{new} \propto (2.25)^3 = 11.391$$

$$\% \text{ Reduction} = \frac{15.625 - 11.391}{15.625} \times 100 \approx \mathbf{27.1}\%$$

5. Safety Analysis (Mixing Tank)

(a) **Material Buckling:**

$$B_m^2 = \frac{1.25 - 1}{L^2} = \frac{0.25}{(6.0)^2} = \frac{0.25}{36} = \mathbf{0.006944} \text{ cm}^{\mathbf{-2}}$$

(b) **Radial Buckling (B_r^2):**

$$B_r^2 = \left(\frac{2.405}{R}\right)^2 = \left(\frac{2.405}{30}\right)^2 = (0.08017)^2 = \mathbf{0.006427}\ \mathrm{cm}^{-2}$$

(c) **Critical Height (H_c):** For criticality, Total Buckling = Material Buckling.

$$B_m^2 = B_r^2 + B_z^2$$

$$B_z^2 = B_m^2 - B_r^2 = 0.006944 - 0.006427 = 0.000517\ \mathrm{cm}^{-2}$$

Since $B_z = \frac{\pi}{H_c}$:

$$H_c = \frac{\pi}{\sqrt{0.000517}} = \frac{\pi}{0.0227} \approx \mathbf{138.2}\ \mathrm{cm}$$

(d) **Status of Tank:** The actual height is $H = 50$ cm. Since $H < H_c$ ($50 < 138.2$) , the system is too short to sustain a reaction. Leakage is too high. **Status: Sub-critical.**

(e) **Safety Check (Increasing R to 50 cm):** If R increases, the Radial Buckling decreases (less leakage out the sides).

$$B_r^2(new) = \left(\frac{2.405}{50}\right)^2 = 0.00231\ \mathrm{cm}^{-2}$$

$$B_z^2(required) = 0.006944 - 0.00231 = 0.00463\ \mathrm{cm}^{-2}$$

$$H_c(new) = \frac{\pi}{\sqrt{0.00463}} \approx \mathbf{46.1}\ \mathrm{cm}$$

Analysis: The critical height *decreases* drastically (from 138 cm to 46 cm). Physically, because the tank is wider, neutrons are less likely to leak out the sides. Therefore, the tank doesn't need to be as tall to keep the neutrons inside. This makes the tank **more dangerous**—a 50 cm fill level would now be Super-critical!